Homework 4
Computation Intelligence and its Applications in Mechatronics
Amirkabir University of Technology

Amirkabir University of Technology
(Tehran Polytechnic)

Submission deadline: **May 10, 12:00 AM**

1 RNN for Sentiment Analysis

In this task, you will build and train a Recurrent Neural Network (RNN) for sentiment analysis on the IMDB dataset. The goal is to classify movie reviews as positive or negative.

1.1 Dataset: IMDB

The IMDB dataset consists of 50,000 movie reviews, split into 25,000 for training and 25,000 for testing. Each set contains an equal number of positive and negative reviews. The reviews have been preprocessed, and each review is represented as a sequence of integers, where each integer signifies a specific word in a dictionary. Load the dataset using TensorFlow/Keras:

```
from tensorflow.keras.datasets import imdb
x_train = pad_sequences(x_train, maxlen=maxlen)
x_test = pad_sequences(x_test, maxlen=maxlen)
```

1.2 Your Tasks

1. Preprocess the loaded data.
2. Embed the words in the sequences.
3. Build an RNN model for sentiment analysis. You are free to design any RNN-based architecture, and you can experiment with different types of RNN layers, such as SimpleRNN, GRU, or LSTM.
4. Compile and train your model on the training set.
5. Evaluate your trained model on the test set and report the test accuracy.
6. Provide a summary of your experiments, the best-performing model configuration, and its test accuracy.

2 Transfer Learning for Sentiment Analysis with DistilBERT

In this task, you will extend your RNN-based sentiment classifier using a pre-trained model: DistilBERT from the KerasNLP library. DistilBERT is a smaller, faster version of BERT that retains 97% of BERT's language understanding while being 60% faster and 40% smaller. You will use the pre-trained DistilBERT from KerasNLP and fine-tune it for binary sentiment classification on the IMDB dataset.

Your Tasks

1. Load the pre-trained DistilBERT model.
2. Fine-tune the model on the IMDB training data.
3. Evaluate the model on the test data and report the accuracy.
4. Compare the performance with your RNN model from Task 1. Highlight the differences in accuracy, training time, and generalization.

Note

- You are not allowed to use any Hugging Face Transformers. Use only KerasNLP APIs.
- You do not need to write a custom BERT architecture. Use the KerasNLP-provided classifier directly.

3 Zero-Shot Learning with CLIP on CIFAR-10

In this task, you will explore the capabilities of zero-shot learning using the CLIP model on the CIFAR-10 dataset. You will evaluate CLIP's accuracy without any specific training on CIFAR-10 by formulating different text prompts for each class.

3.1 Your Tasks

1. Load the CIFAR-10 test dataset.
2. Load a pre-trained CLIP model.
3. For each of the 10 classes in CIFAR-10, formulate several different text prompts.
4. Calculate and report the zero-shot classification accuracy on the CIFAR-10 test set for each set of prompts you designed.
5. Analyze how different prompts affect the zero-shot accuracy. Identify the prompt or set of prompts that yields the best performance.

Extra Point

- The group that achieves the highest test accuracy for the first task among all submissions will receive an extra point for this assignment.
- Implementing task 2 using Hugging Face Transformers and comparing its performance and training behavior with the KerasNLP model.

Submission Guidelines

- Homeworks must be completed in **groups of four**.
- Each group must submit a single file containing a **PDF report**, including answers to discussion questions, visualizations, and code explanations, and a **Jupyter Notebook**, containing all code and results.